Physical examination of sheep and goat

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Objectives

- Basic physical examination
- Detect abnormalities instead of memorize the diseases
- Do no harm

Signalment

History

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Large mammals Form filler: _____

ID: _____ Date: _____

Common name _____

Patient name _____

Date Acquisition: _____

Place acquisition / previous owner: _____

Sex: ♀ ☐ ♂ ☐ Unknown ☐ Age: _____ BW: _____

CC: _____

Environment

Ambient temperature _____

Light source and cycle _____

Ventilation _____

Stock density / companion _____

Flooring _____

bedding _____

Cleaning _____

Diet

Water / food location _____

Other livestock accessibility _____

Diet proportions / supplements / frequency _____

_____

_____

History

Vaccination _____ anthelmintics _____

Previous disease problem _____

Chief complaint _____

_____

_____

Duration _____

progression _____

Treatment already _____

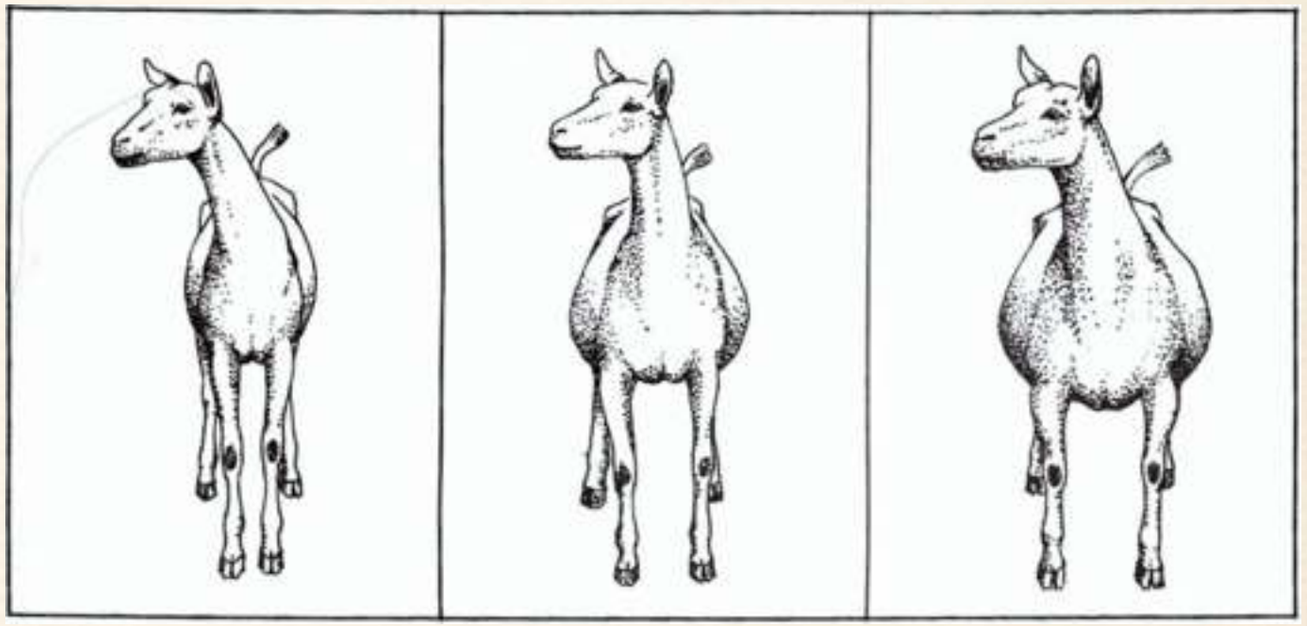
Recent change (diet / staff) _____

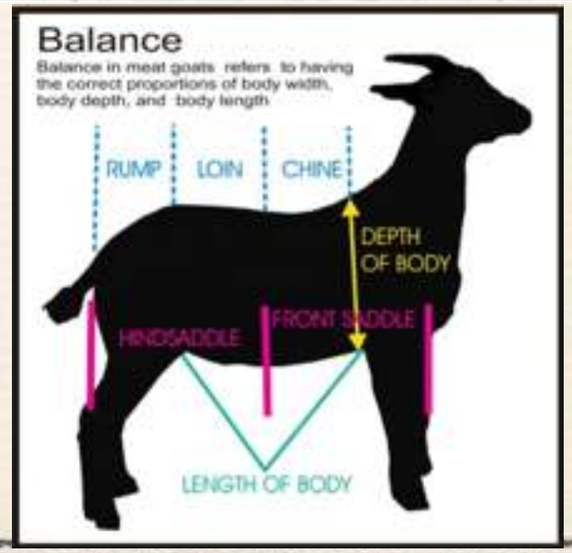
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Distance examination

- Interaction with environment and herd mates
- Gait
- Balance
- Hair coat

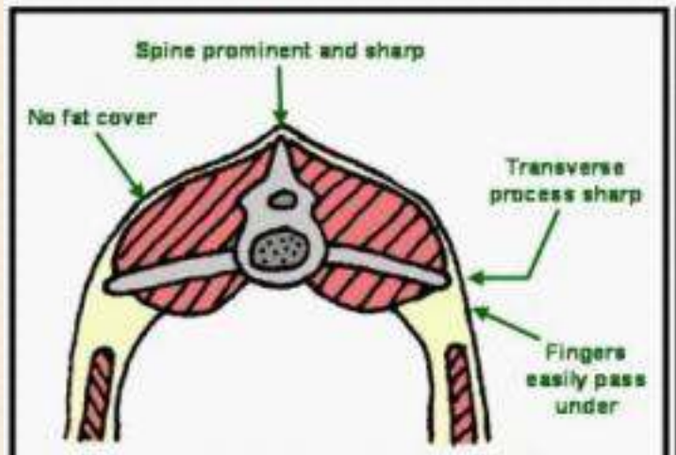




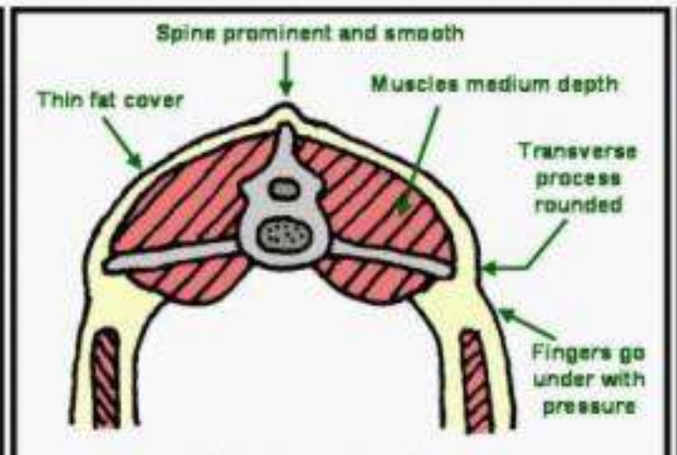
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Body condition score

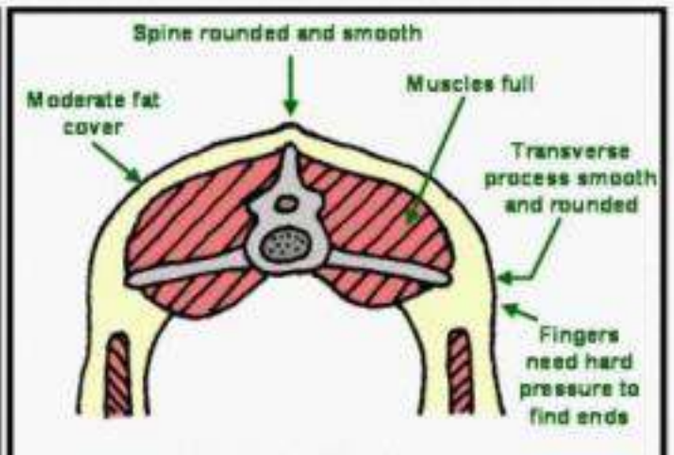
Body Condition Scores – Sheep/Goats



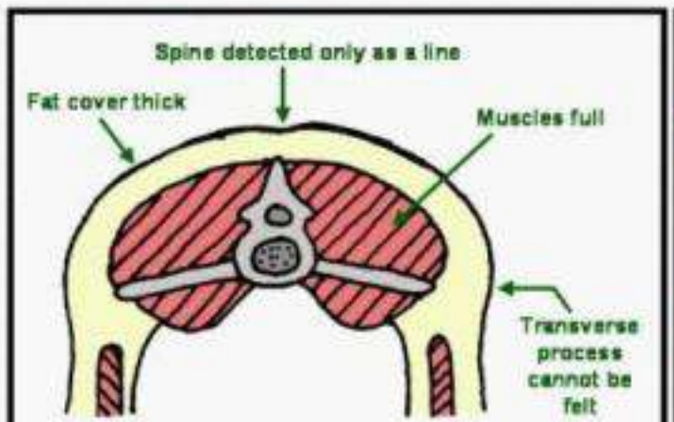
Condition 1 - *Emaciated*



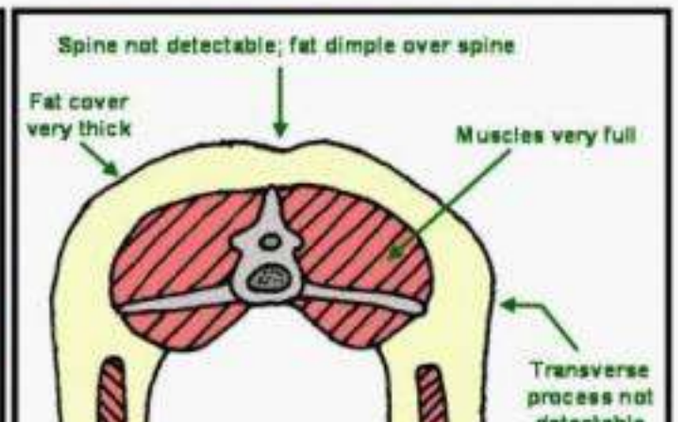
Condition 2 - *Thin*



Condition 3 - *Average*



Condition 4 - *Fat*



Condition 5 - *Obese*

Adapted from "Body Condition Scoring of Sheep" by J.M. Thompson and H. Meyer (Oregon State University)

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Body condition score



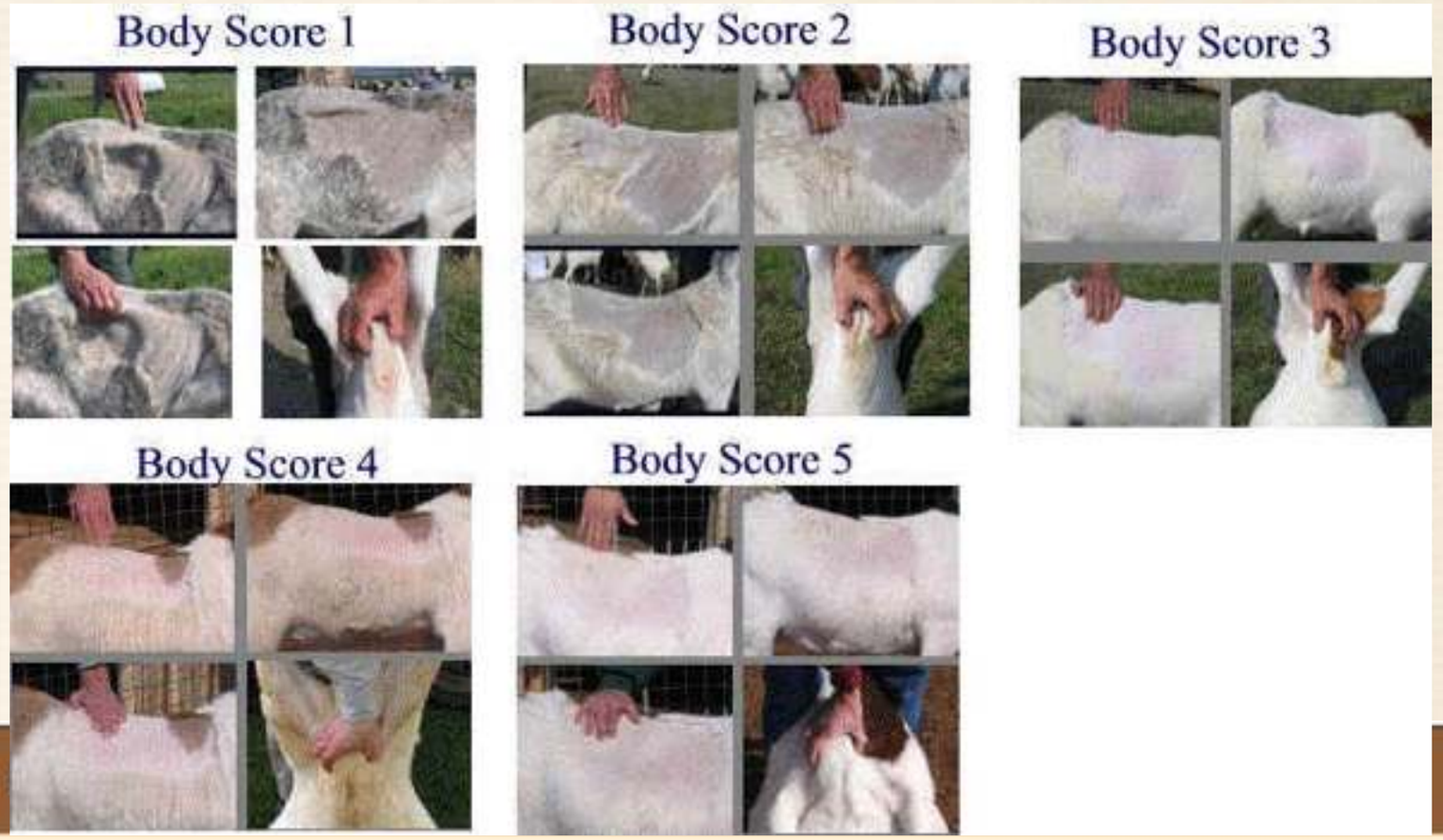


TABLE 1-1 Body Condition Scoring

Assigned Score	Physical Finding: Spinous Processes	Transverse Processes	Loin Eye Muscle	Fat Cover Over Loin Eye Muscle
Condition 1	Sharp and prominent	Sharp	Shallow	None
Condition 2	Sharp and prominent	Smooth, slightly rounded	Medium depth	Little
Condition 3	Smooth and rounded	Smooth, well covered	Full	Medium
Condition 4	Palpable as firm line with pressure	Not palpable	Full	Thick
Condition 5	Not palpable	Not palpable	Very full	Very thick

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Head and neck

- General symmetry
  - Cranial nerve deficit
  - Cud or mass retention
  - Swelling under the chin
    - Submandibular edema
      - Hypoproteinemia
        - Parasitism
    - Enlarged lymph node
    - Goiter
    - Salivary gland abscess





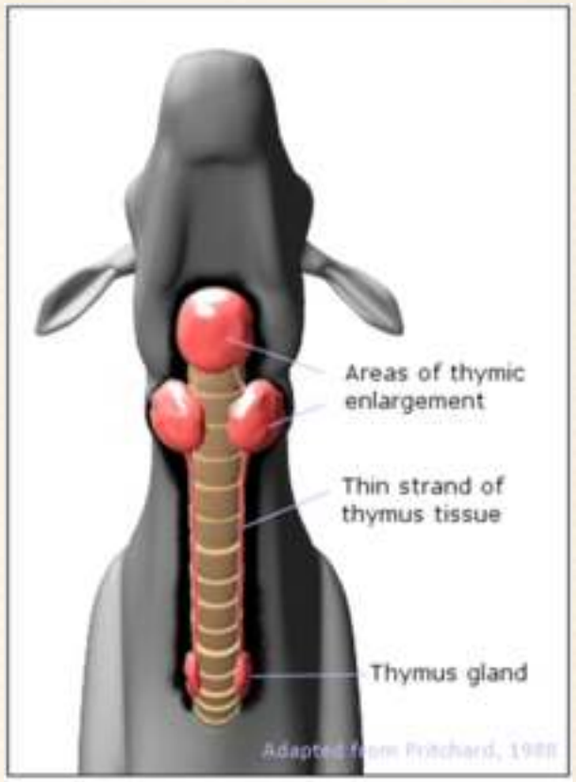


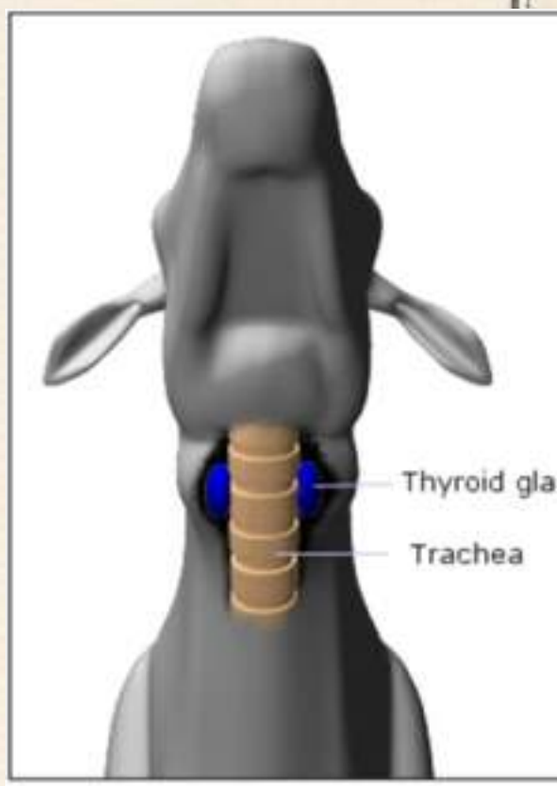
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Head and neck

- Thymus gland
  - Milk gioter
    - Weeks to 4-9months
    - Maturation of immune system
- Goiter
  - Iodine deficiency
  - Soil





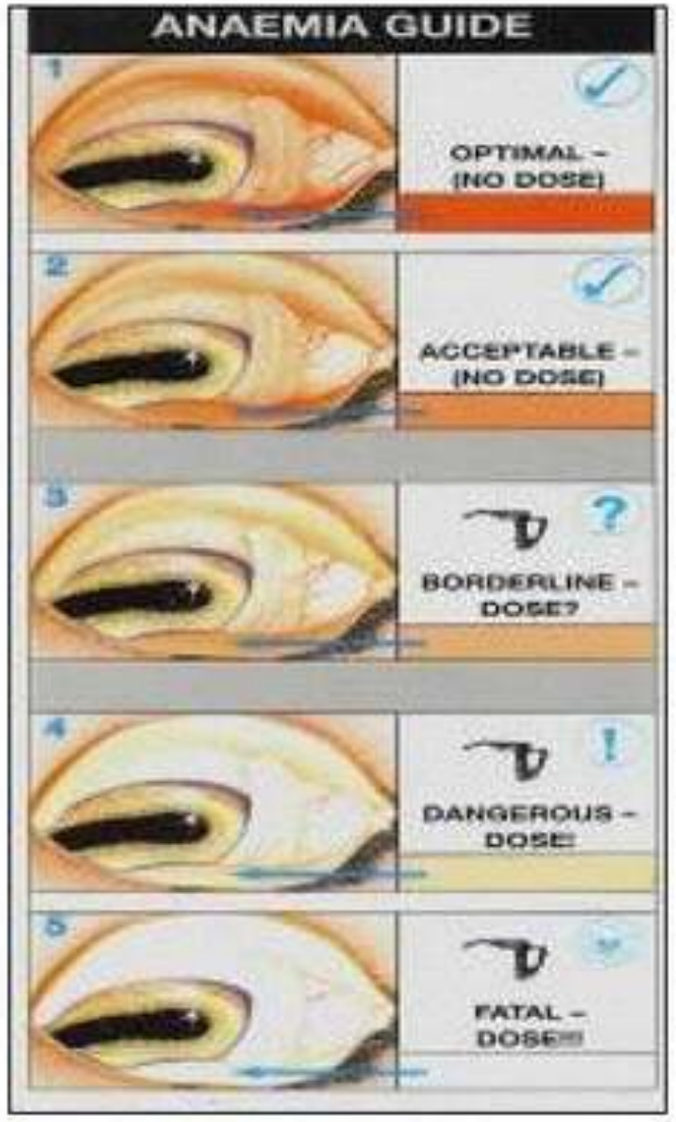


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Eye

- Anemia
  - FAMACHA system
  - Parasite control
- Jaundice
  - Liver disease
  - Hemolytic event
    - Copper toxicity
- Pink eye
  - *Chlamydia psittaci, Chlamydia Ovis*
  - *Mycoplasma conjunctivae*
- Eye abnormalities
- Hydration







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Oral cavity

- Structural
  - Skin & mucus membrane interface
  - Hard palate
  - Vesicles
    - FMD
- Mucus membrane
  - Blue tongue disease









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Oral cavity

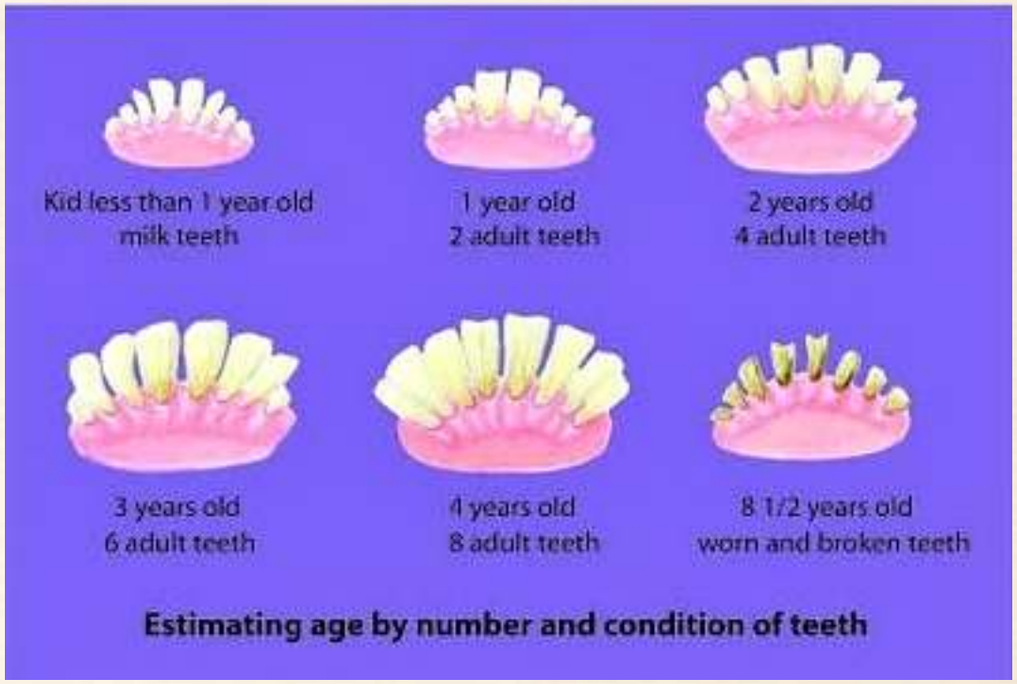
- Odor
  - Infectious diseases
    - Oral cavity
      - Abscessed teeth
      - Rumen
      - Respiratory tract
- Rancid milk odor
  - Neonate with cleft hard palate
- Sweet smell
  - Ketoacidosis

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Dental examination

- Occlusion
- Soft tissue injury







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Auscultation

- Heart
  - Between forelimbs
    - 4-5 rib
- Lung
  - Rib #5 cranially and #11 caudally
  - Under the elbow, in front of the shoulder





TABLE 1-2 Temperature, Pulse, and Respiratory Rates in Adult Sheep and Goats

Parameter	Sheep	Goats
Rectal temperature (° F)	102-103.5	100.5-103.5
Rectal temperature (° C)	39-40	38-40
Pulse (beats/minute)	70-80	70-90
Respiration (breaths/minute)	12-20	15-30

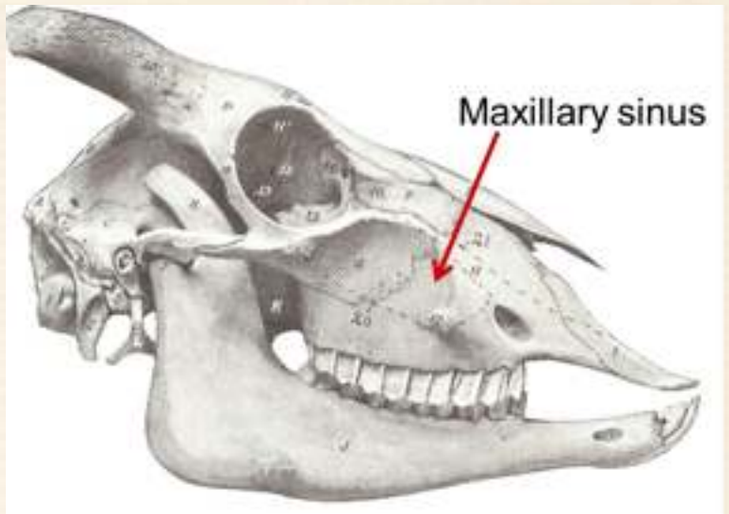
TABLE 1-3 Temperature, Pulse, and Respiratory Rates in Lambs and Kids

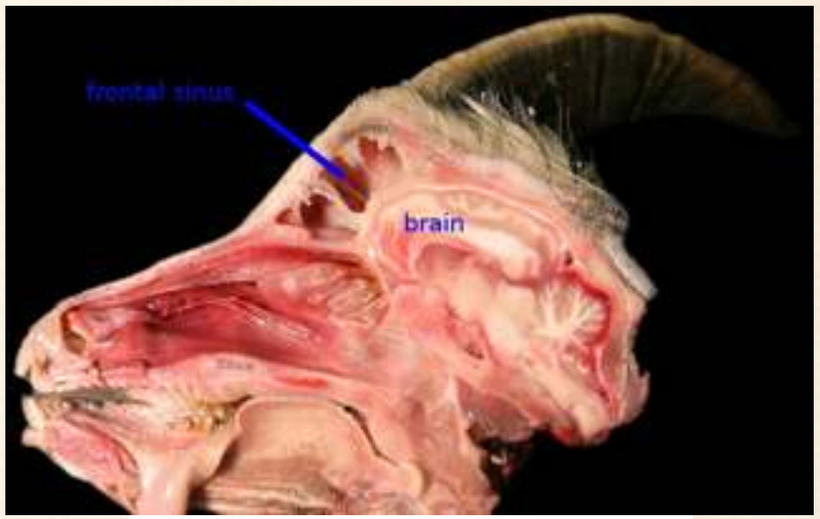
Parameter	Lambs	Kids
Rectal temperature (° F)	102.5-104	102-104
Rectal temperature (° C)	39.5-40.5	39.5-40.5
Pulse (beats/minute)	80-130	90-150
Respiration (breaths/minute)	20-40	20-40

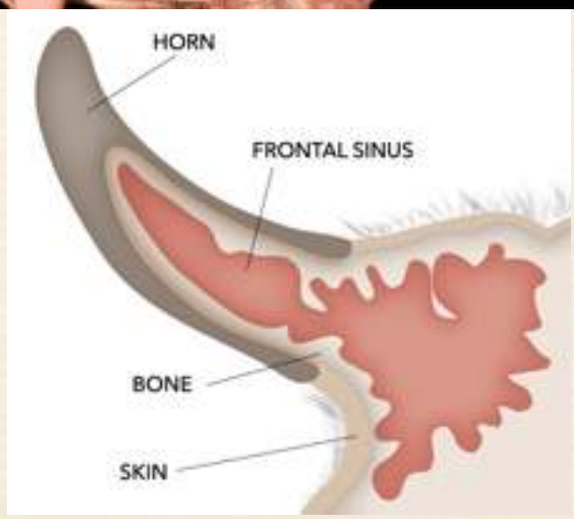
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Auscultation

- Dull sound percussion on sinus
  - Maxillary sinus
    - Tooth root abscess
  - Frontal sinus
    - Infected dehorn site
    - Ascending respiratory infection
- Tumor
  - Rare







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Auscultation

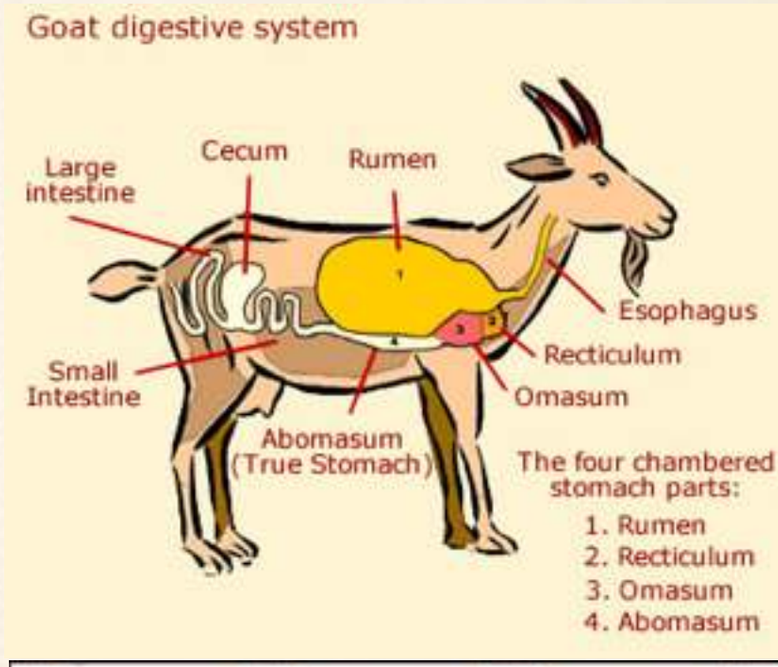
- Trachea
  - Wheeze / Crackle
- Larynx and trachea palpation
  - Repeated cough is abnormal

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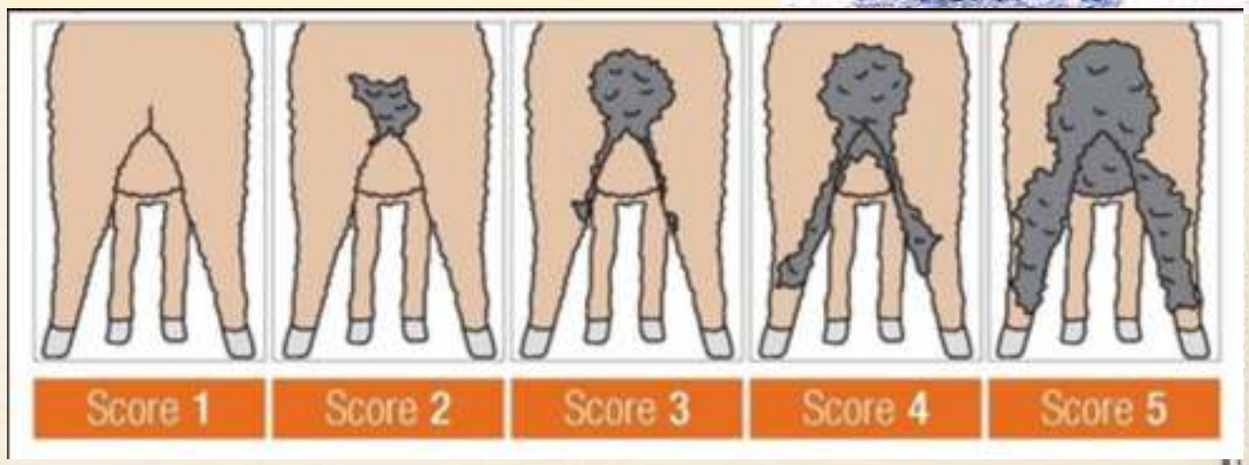
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Gastrointestinal system

- Abdominal contour
  - Left
    - Rumen
      - Mild asymmetry
      - Left paralumbar fossa
        - 1-2 contraction / minute
  - Right
    - Upper right paralumbar fossa
      - Spiral colon
      - Cecum
      - Small intestine
    - Lower right
      - Abomasum
      - Pregnancy
- Generalized
  - Liver, heart disease, parasitism
- Fecal soiling







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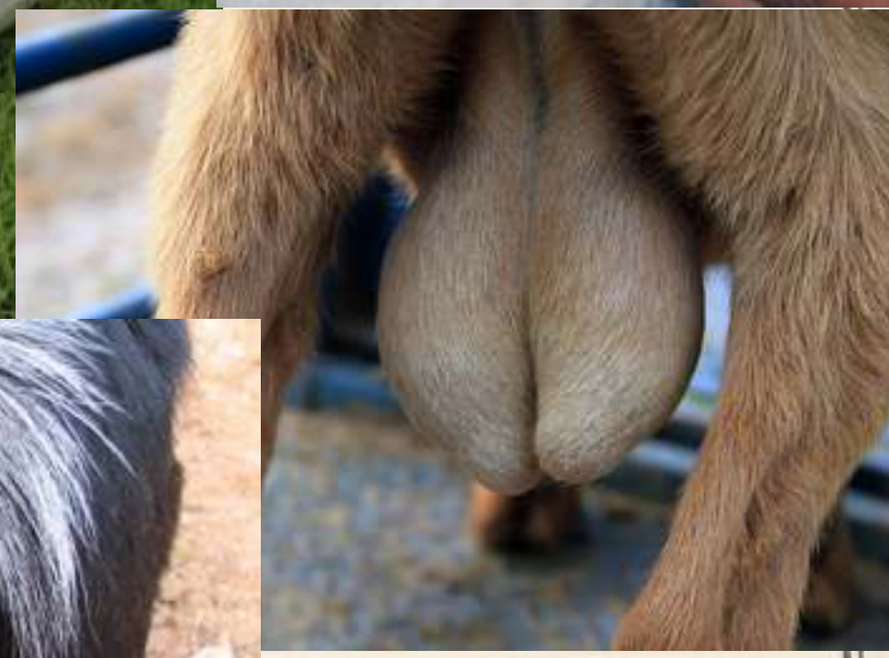
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Urogenital tract

- Prepuce
  - Signs for urolithiasis
    - Lateral recumbency
    - Upright position
- Scrotum
  - Distorsion
- Vulva
  - Discharge
  - Vulvitis







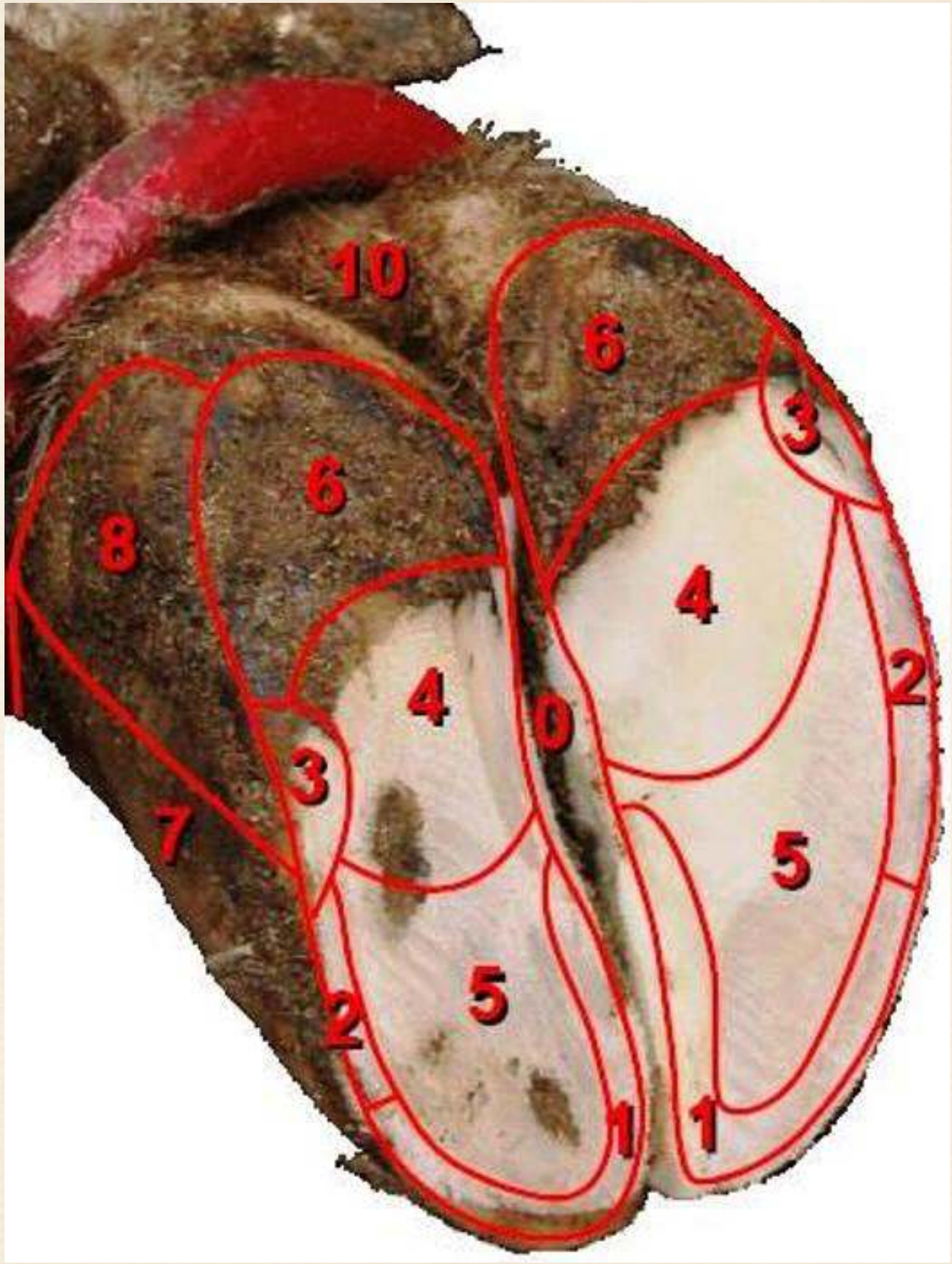




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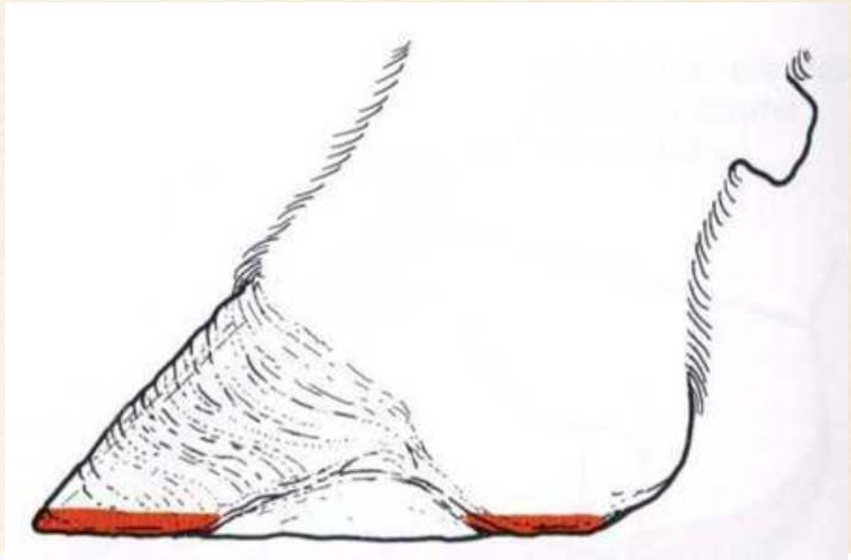
0. Interdigital Region (Skin)
1. White Line (Hoof Wall)
2. White Line (Hoof Wall)
3. White Line (heel bulb Junction)
4. Caudal Aspect of the Sole
5. Anterior Aspect of the Sole
6. Heel Bulb
7. Hoof Wall (Interior ½ of Claw)
8. Hoof Wall (Posterior ½ of Claw)
9. Coronary Band and Skin
10. Skin Above and Between Heels

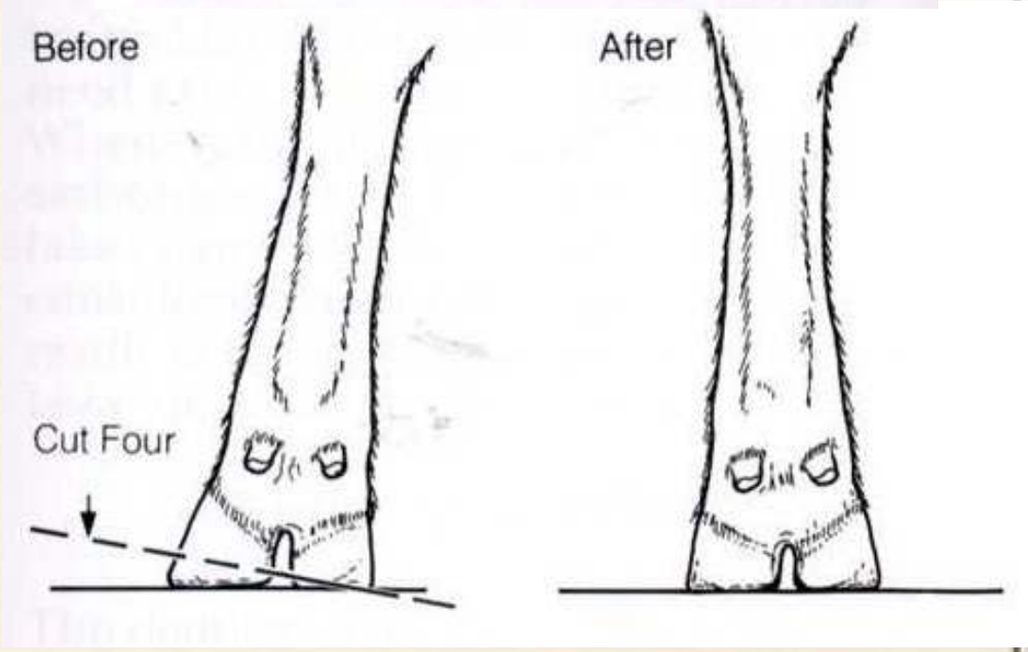
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Musculoskeletal system

- Hoof
  - Not enough wearing
  - Humid environment
  - Foot rot
    - Infection in hoof
    - Contagious
    - Contaminated soil
- Foot scald
  - Interdigit ulcers
  - Precursor of foot rot and foot abscess





Foot scald	Foot rot	Foot abscess











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Musculoskeletal system

- Joint
  - Palpation
    - Heat, pain, swelling
- Range of motion
- CAE
- Hygroma





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Lymphatic system

- Lymph node
  - May not be palpable
- Caseous lymphadentits

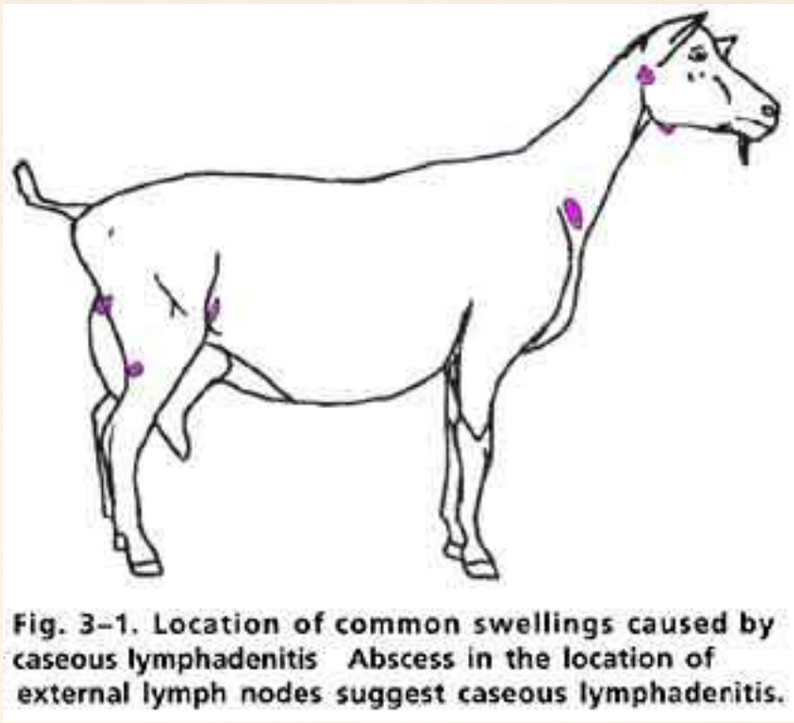


Fig. 3–1. Location of common swellings caused by caseous lymphadenitis Abscess in the location of external lymph nodes suggest caseous lymphaderitis.











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Mammary gland

- Symmetry
  - Size
  - Color
  - Temperature
  - Shape
  - Consistency



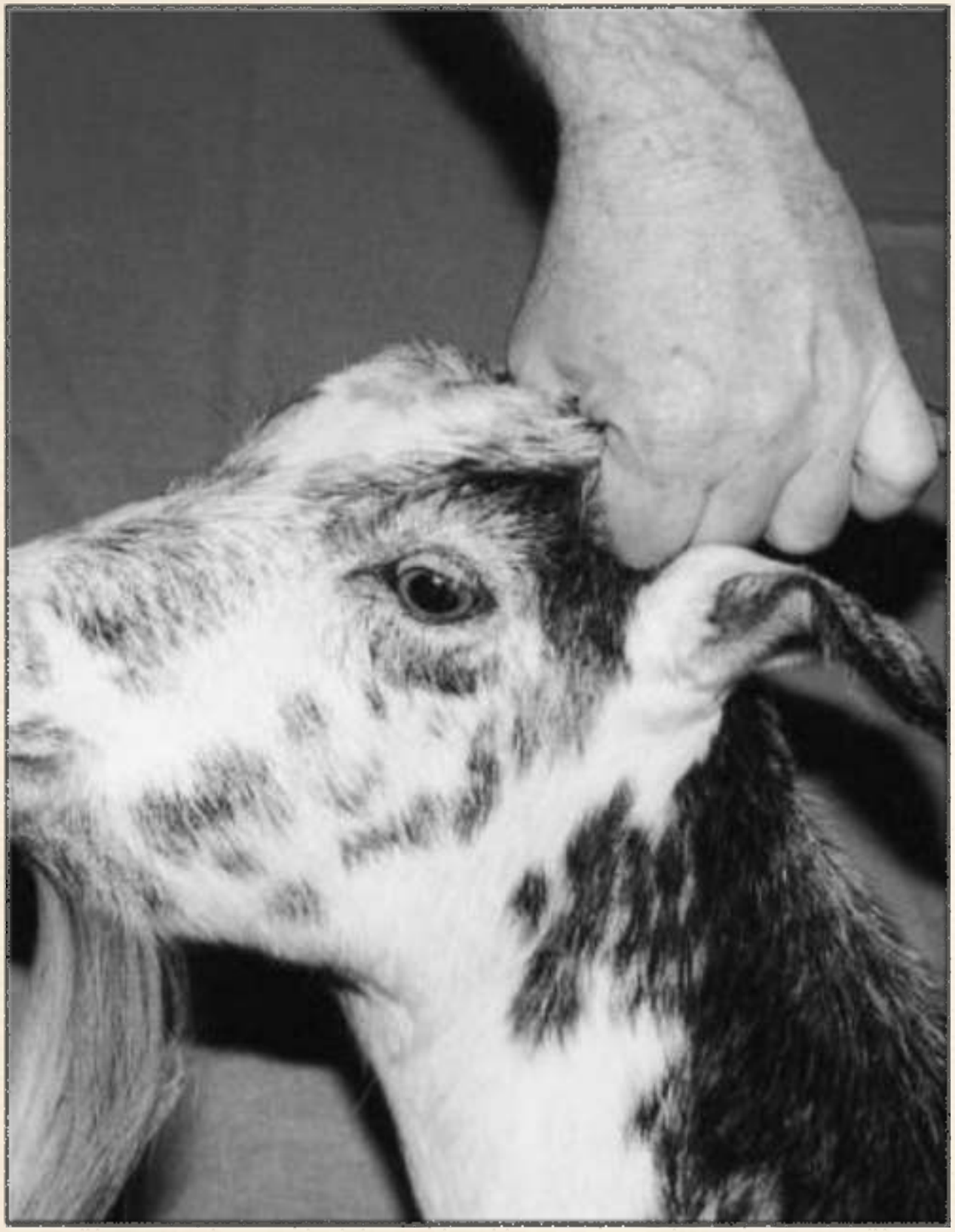




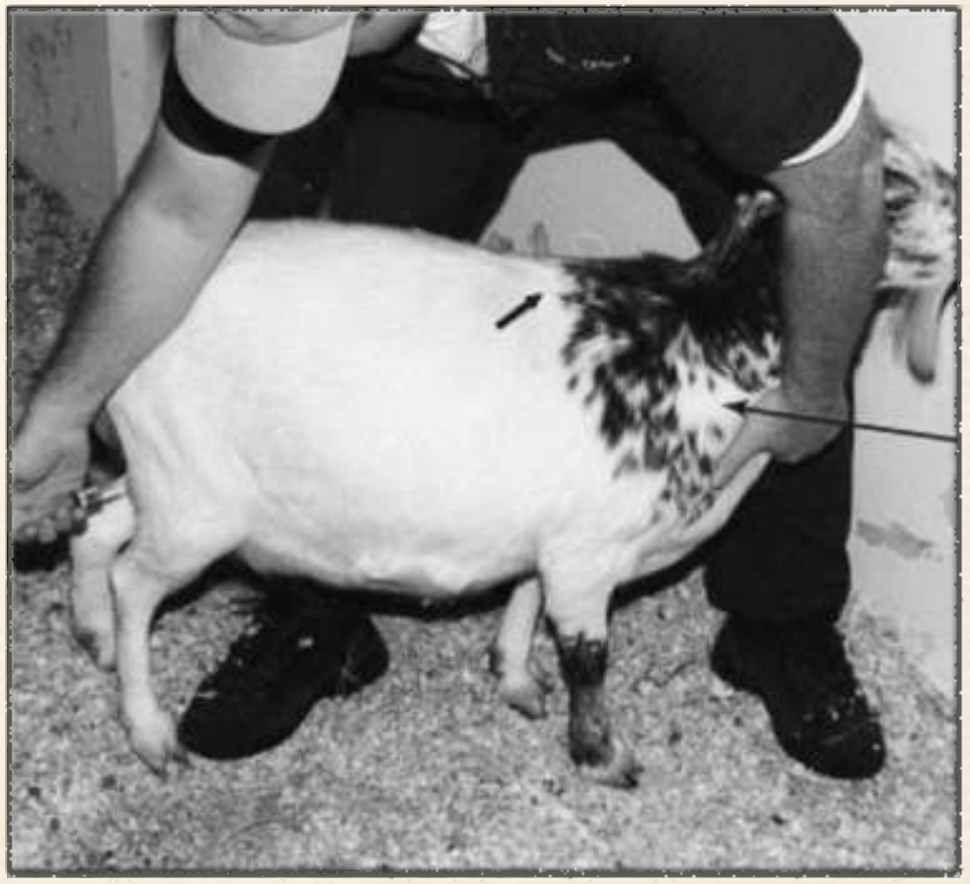




Clinical pathology of goat and sheep







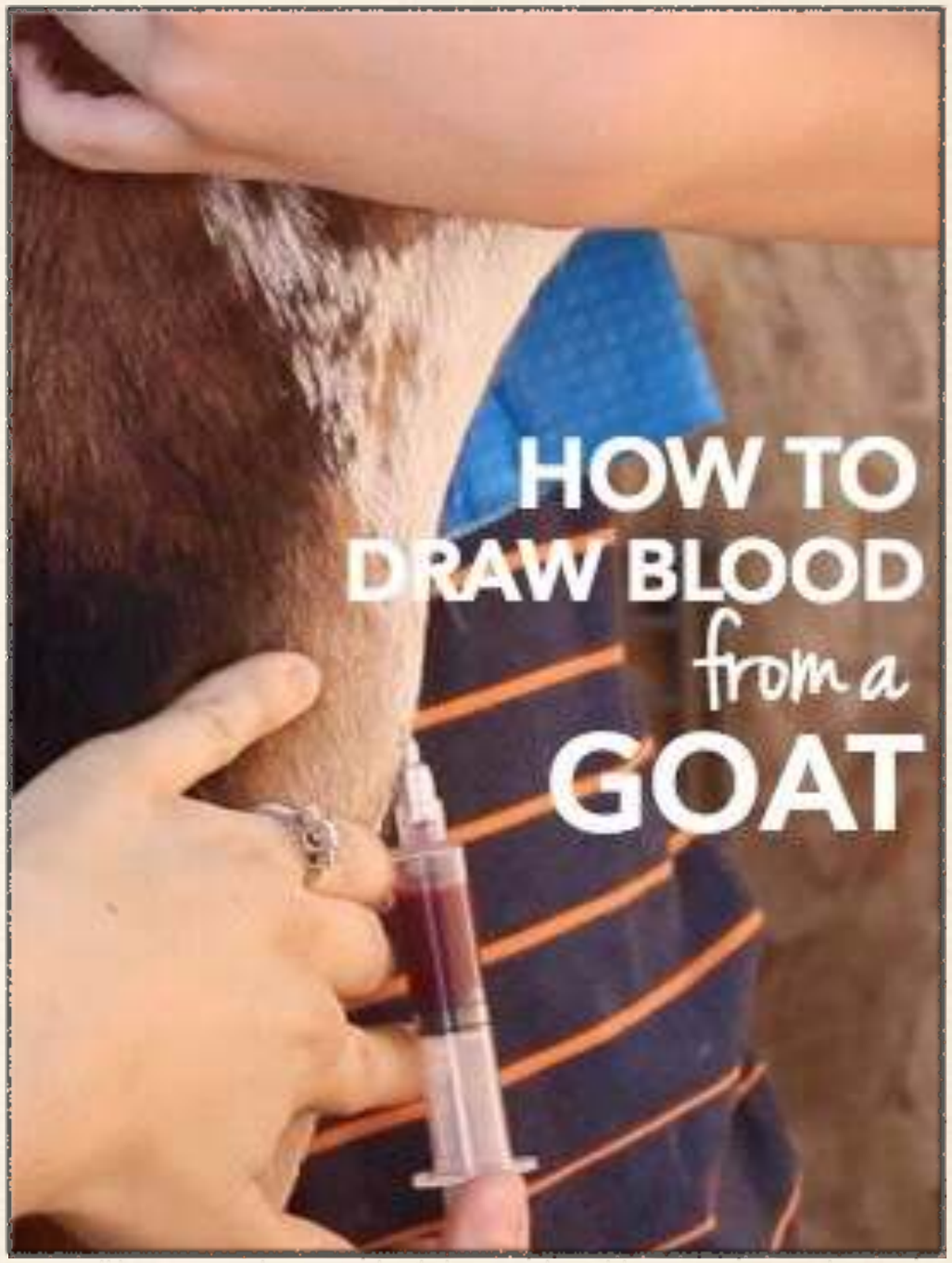









B


HOW TO
DRAW BLOOD
from a
GOAT

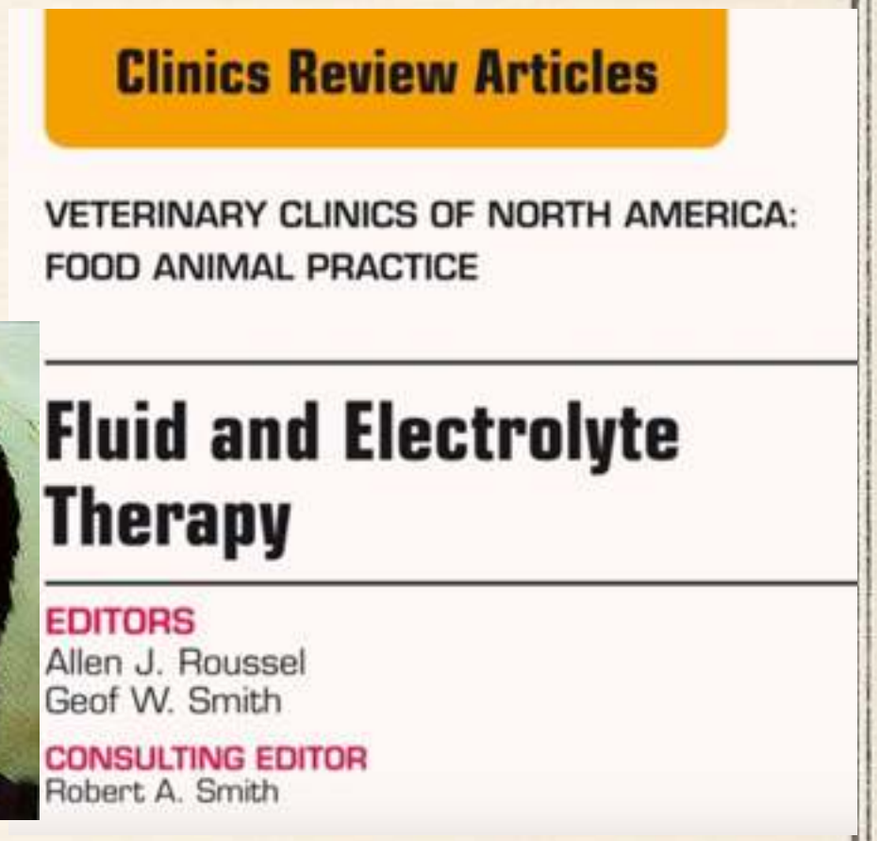
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Fluid and electrolyte balance

- Total body water
  - Intracellular fluid
    - 40% body weight
  - Extracellular fluid
    - 20% body weight
      - Intravascular fluid
        - 5% BW
      - Interstitial fluid
        - 15% BW
          - Fluid surrounding cells, CSF, connective tissues
          - Fluid in reticulorumen and GI tract





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Fluid and electrolyte balance

- Equilibrium between ECF and ICF
  - Fluid
    - Hydrostatic
    - Oncotic
- Electrolyte
  - Na^+-K^+ ATPase pump
  - Chloride, bicarbonate vs. phosphate, protein

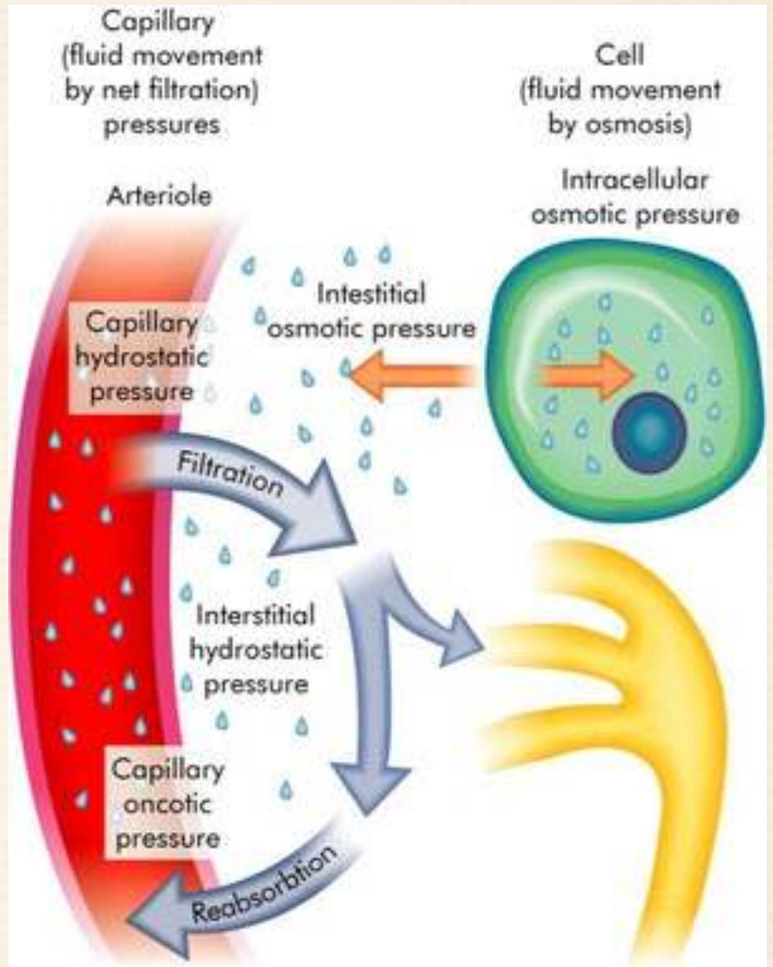


Fig. 4-1. Fluid Movement Between Plasma and Interstitial Space
The movement of fluid between the vascular, interstitial spaces, and the lymphatics is the result of net filtration of fluid across the semipermeable capillary membrane. *Capillary hydrostatic pressure* is the primary force for fluid movement out of the arteriolar end of the capillary and into the interstitial space. At the venous end, *capillary oncotic pressure* (from plasma proteins) attracts water back into the vascular space. *Interstitial hydrostatic pressure* promotes the movement of fluid and proteins in to the lymphatics. *Osmotic pressure* accounts for the movement of fluid between the interstitial space and the intracellular space. Normally intracellular and extracellular fluid osmotic pressures are equal (280 to 294 mOsm) and water is equally distributed between the interstitial and intracellular compartments.

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Fluid and electrolyte balance

- Dehydration
  - Common feature of many diseases
    - Concurrent fluid and electrolyte loss
      - Hypernatremia/Hyperchloridemia not common
    - Acid - base abnormality
- Laboratory testing aid needed

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Fluid and electrolyte balance

- Sodium
  - Hyponatremia
    - Water excess
    - Hypotonic dehydration
      - Chronic dehydration
  - Hypernatremia
    - Water deficit
    - Hypertonic dehydration
  - Unchanged
    - Isotonic dehydration
      - Acute diarrhea, heavy sweating

$$\text{Serum Na(mEq/L)} \simeq \frac{\text{Exchangeable(Na + K)}}{\text{Total body water}}$$

BOX 22-2

Causes of Hypernatremia

COMMON CAUSES
Pure water losses
- Panting
- Water deprivation

Sodium excess (water restriction)
- Salt poisoning
- Feeding only electrolytes, no free water

UNCOMMON CAUSES
Water loss exceeds electrolyte loss
- Vomiting
- Diarrhea
- Burns
- Intrinsic renal disease
- Diuretics
- Diabetes insipidus
  - Central
  - Nephrogenic
- Hypertonic saline or sodium bicarbonate administration
- Mineralocorticoid excess

BOX 22-1

Causes of Hyponatremia

COMMON CAUSES
Relative water excess
- Loss of sodium-containing fluid (decreased effective circulating volume)
  - Diarrhea
- Excessive sweating
- Blood loss
- Fluid drainage
  - High-volume gastric reflux
  - High-volume pleural drainage

Adrenal insufficiency
Sequestration of fluid (third-space problems)
- Peritonitis
- Ascites
- Ruptured bladder
- Torsion or volvulus of the gut
- Excessive administration of 5% dextrose to patient with renal disease

False hyponatremia
- Hyperlipidemia
- Hyperproteinemia
- Hyperglycemia

UNCOMMON CAUSES
Water retention with normal effective circulating volume
- Psychogenic polydipsia
- Renal disease
- Inappropriate antidiuretic hormone secretion

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Fluid and electrolyte balance

- Potassium
  - Dehydration
  - Acid-base imbalance
  - Renal function
  - Urinary tract obstruction
- Chloride
  - Proportional change in sodium concentration
  - Vary inversely with bicarbonate concentration

Monitor

$$CO_2 + H_2O \leftrightarrow H^+ + HCO_3^-$$

- HCO_3
  - Major acid buffering compound in the body
  - Altered value
    - Altered pH value
    - Activated compensatory mechanism

TABLE 1 VETERINARY MEDICINE

Normal Values and Ranges for Arterial Blood Gas Analysis in Dogs, Cats, and People*

	Dogs	Cats	People
$PaCO_2$ (mm Hg)	36.8 (30.8–42.8)	31.0 (25.2–36.8)	40 (35–45)
PaO_2 (mmol/L)	92.1 (80.9–103.3)	106.8 (95.4–118.2)	97 (> 80)
HCO_3^- (mmol/L)	22.2 (18.8–25.6)	18.0 (14.4–21.6)	24 (22–26)
pH	7.407 (7.351–7.463)	7.386 (7.310–7.462)	7.40 (7.35–7.45)

*Adapted from Haskins SC. Blood gases and acid-base balance: clinical interpretation and therapeutic implications. In Kirk RW, ed. *Current veterinary therapy VIII small animal practice*. Philadelphia, Pa: WB Saunders, 1983; 201.

Monitor

- Acid base disturbance
  - Corrected before anesthesia
  - Tolerated during anesthesia
  - Corrected slowly
    - Unless
      - Respiratory event
- Step. 1
  - pH value
- Step. 2
  - Respiratory or Metabolic
- Step. 3
  - Compensatory response
    - Duration

Characteristics of Primary Acid-Base Disturbances*

Disorder	Primary Disturbance	Compensatory Response
Respiratory acidosis	Increased $PaCO_2$	Increased HCO_3^-
Respiratory alkalosis	Decreased $PaCO_2$	Decreased HCO_3^-
Metabolic acidosis	Decreased HCO_3^-	Decreased $PaCO_2$
Metabolic alkalosis	Increased HCO_3^-	Increased $PaCO_2$

*Adapted from DiBartola SP. Introduction to acid-base disorders. In: *Fluid, electrolyte and acid-base disorders in small animal practice*. 3rd ed. St. Lou
Mo: Saunders Elsevier, 2006;236.

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Fluid and electrolyte balance

- Base deficit / Base deficit
  - Provided value in routine acid-base assessment
  - Amount of bicarbonate needed to correct metabolic acidosis
    - Neonates : mEq bicarbonate needed
      = (base deficit) x (body weight in kg) x 0.6
    - Adults : mEq bicarbonate needed
      = (base deficit) x (body weight in kg) x 0.3

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Fluid and electrolyte balance

- Anion gap
  - Help Determine causal factor of acid-base imbalance
  - $AG=Na^{+}+K^{+}-Cl^{-}-HCO3^{-}$
    - Major ions-measured anions
    - 12-16
  - Approximation of unmeasured anions
    - Protein(albumin), sulphate, phosphate, lactate, ketone
    - Species difference
- Decreased AG acidosis
  - Hypoalbuminemia, hyperchloridemia
    - With hypokalemia
      - Diarrhea
    - With hyperkalemia
      - Renal failure

Effect of serum strong ions on metabolic acid-base balance

Independent strong ion	↑ Concentration	↓ Concentration
Sodium concentration (water balance)	Alkalosis	Acidosis
Relative chloride concentration	Acidosis	Alkalosis
Protein concentration	Acidosis	Alkalosis
Unmeasured anions (eg, lactate, ketoacids, phosphate, sulfate)	Acidosis	Alkalosis

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Fluid and electrolyte balance

- Anion gap
  - Increased AG
    - Increased metabolized acid
      - Lactate
        - Anaerobic exercise
        - Grain overload
        - Hypovolemic shock
    - Ketonacidosis
    - Uremic acidosis

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Fluid and electrolyte balance

- Calcium
  - Nerve impulse, muscle contraction, blood clotting
    - 6 mg/dl recumbent
    - 4 mg/dl fatality
  - Common metabolic regulation
  - Protein and ionized calcium
    - Acidosis unbound calcium and proteins
- Phosphorus
  - Dietary, hormones, vit D
- Magnesium
  - Nutrition and environmental interaction

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Fluid and electrolyte balance



- Parturient paresis
  - Hypocalcemia, hypophosphatemia, normal magnesium
    - Goat
      - Lambing
    - Sheep
      - 1 month before lambing



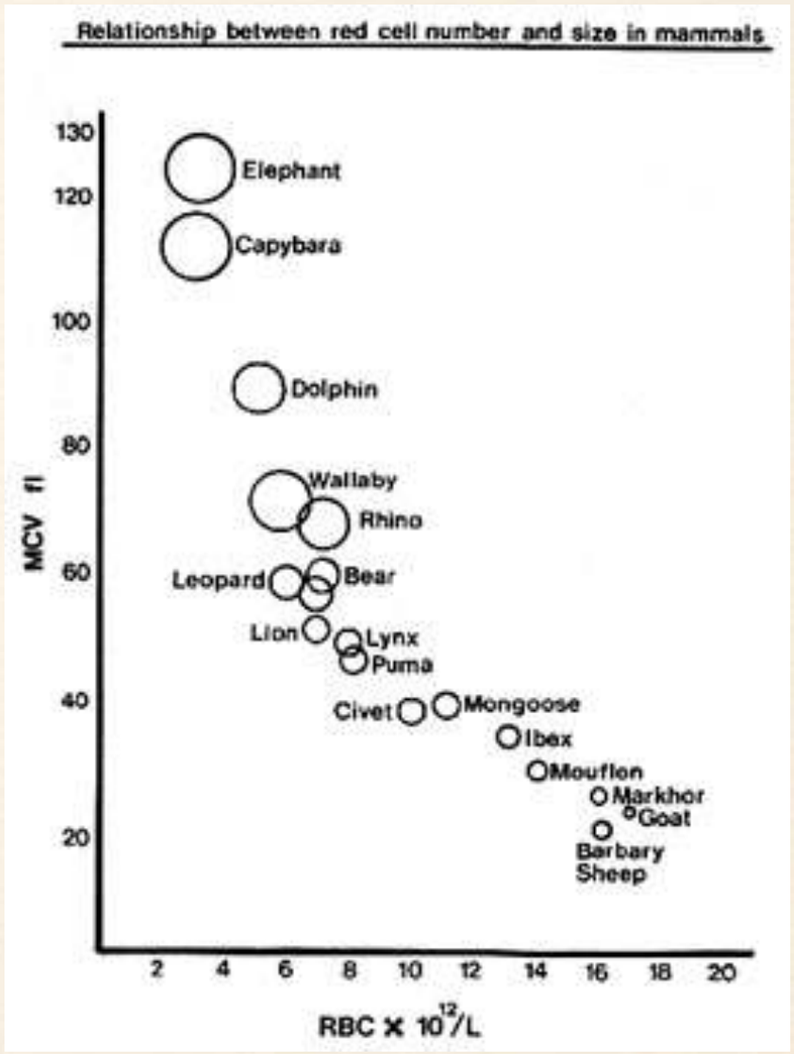
- Grass tetany
  - Hypomagnesium, hypocalcemia, normal calcium

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Hematology

- Goat erythrocyte
  - Small
    - Prone to hemolysis
      - Reported as platelet in machine
      - Vacutainer prohibited
    - EDTA tube
    - Morphology provide important information
- WBC
  - Inflammatory response
    - Degenerate, toxic, immature
    - No change in WBC count



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Hematology

- Anemia
  - Blood loss
    - Trauma
    - Parasite
  - Hemolysis
    - Toxin
      - Copper
        - Sheep
      - Onion
      - Bracken fern
        - Heinz body
          - Oxidative damage



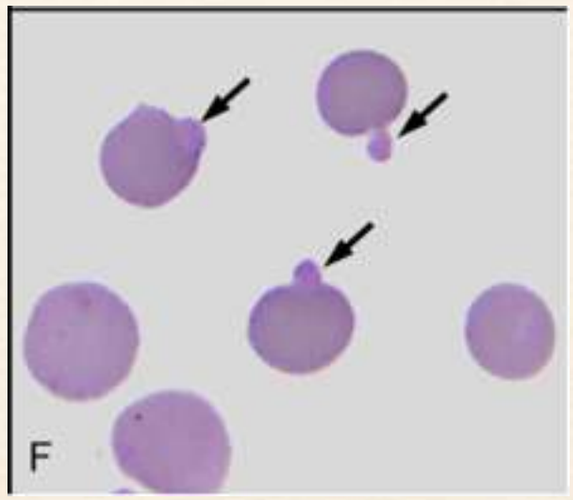


TABLE 16-1 Normal Hematologic Parameters For Sheep And Goats

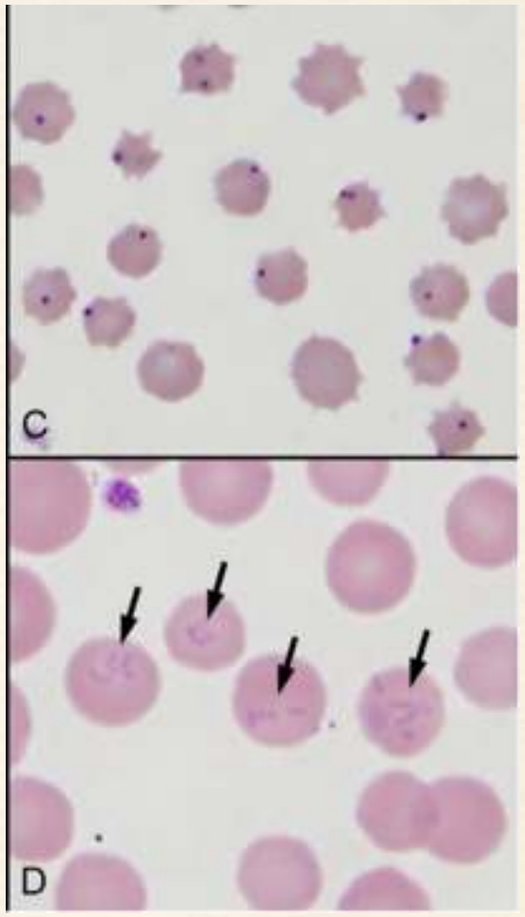
Parameter (Units)	Adult Sheep	Adult Goat
Hematocrit (%)	27-45	22-36
Hemoglobin (g/dL)	9-15.8	8-12
Red blood cell count ($\times 10^6/\mu L$)	9-17.5	8-17
Mean corpuscular volume (fL)	28-40	15-26
Mean corpuscular hemoglobin concentration (g/dL)	31-34	29-35
Platelet count ($\times 10^5/\mu L$)	2.4-7.0	2.8-6.4
Total white blood cell count (/μL)	4000-12,000	4000-13,000
Segmented neutrophils (/μL)	1500-9000	1400-8000
Band neutrophils (/μL)	0	0
Lymphocytes (/μL)	2000-9000	2000-9000
Monocytes (/μL)	0-600	0-500
Eosinophils (/μL)	0-1000	0-900
Basophils (/μL)	0-300	0-100
Total plasma protein (g/dL)	6.2-7.5	6.0-7.5
Fibrinogen (mg/dL)	100-600	100-500

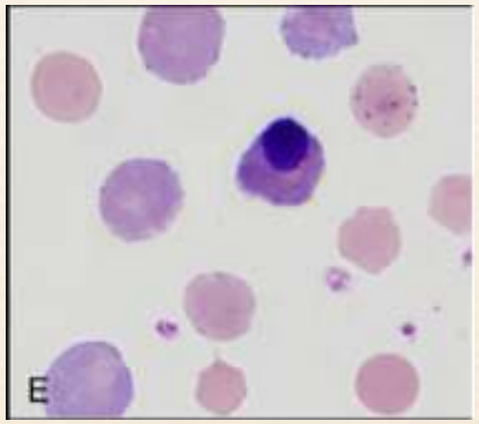
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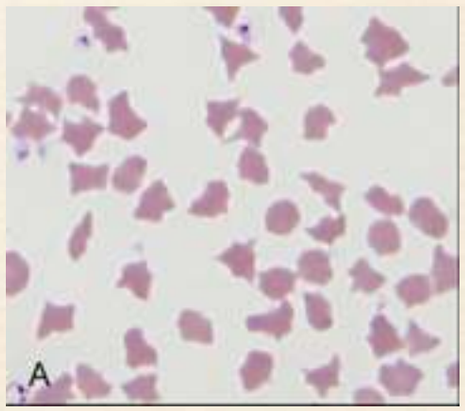
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Hematology

- Anemia
  - Hemolysis
    - Immune mediated
      - Rare
      - Penicillin
      - Bovine colostrum to small ruminant
    - RBC parasitism
    - Bacterial toxin
      - Clostridium
      - Leptospira
  - Chronic disease
    - Pneumonia, footrot, caseous lymphadenitis,
    - Malnutrition,
    - Environmental stressors
  - Young calf
    - Poikilocyte





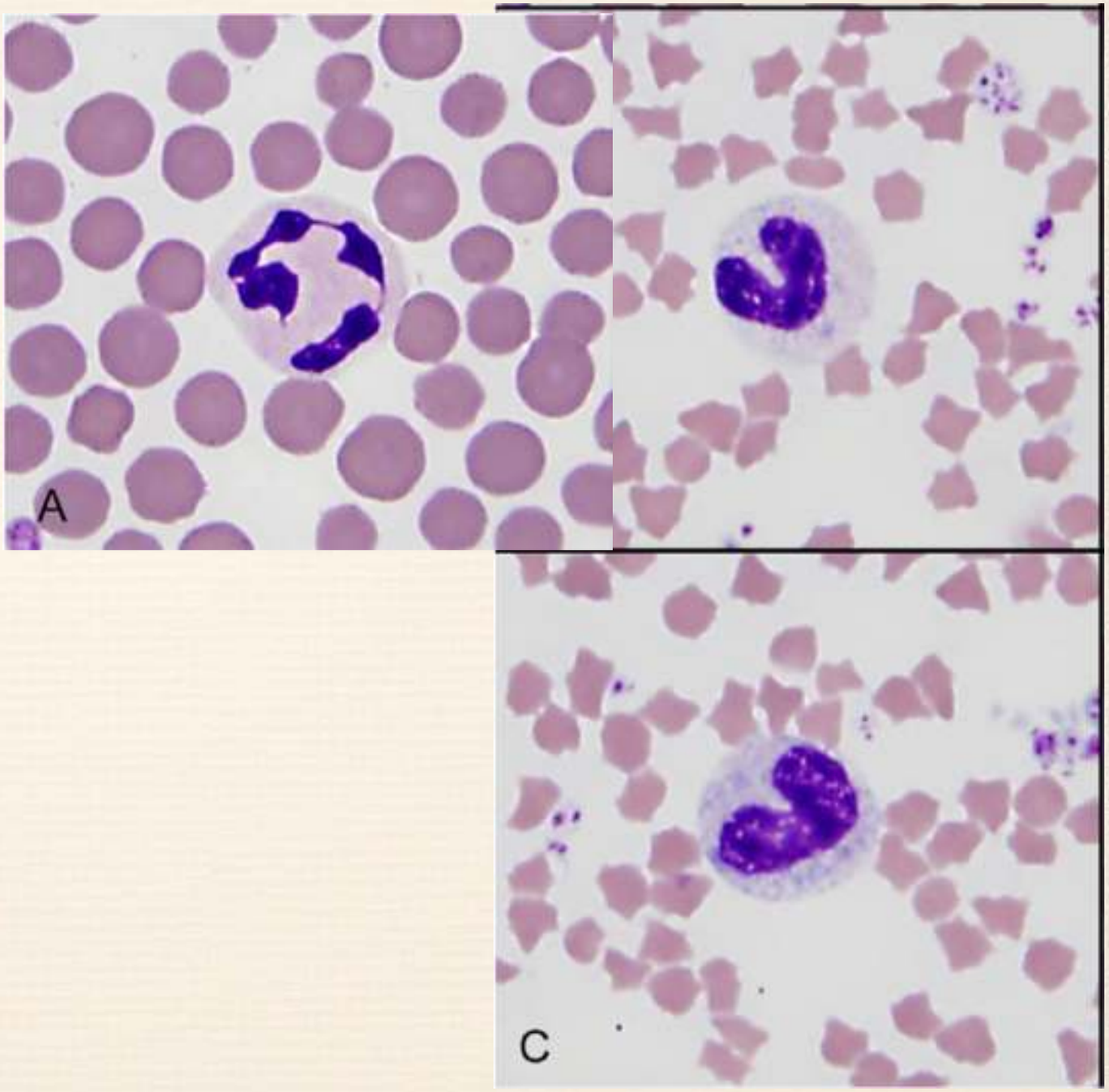


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Hematology

- Neutrophil
  - Dominate in young ruminant
  - 1:1 or 3:2 in goat > 3years old
- Inflammatory leukogram
  - Change in neutrophil proportion
  - Immature neutrophil
    - Band cell
- Stress leukogram
  - Neutrophilic, lymphopenia, eosinopenia
  - No left shift

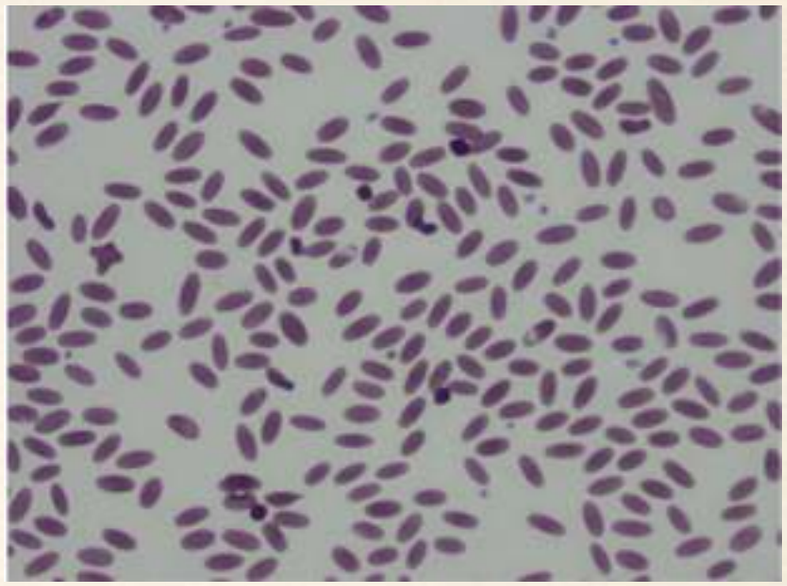


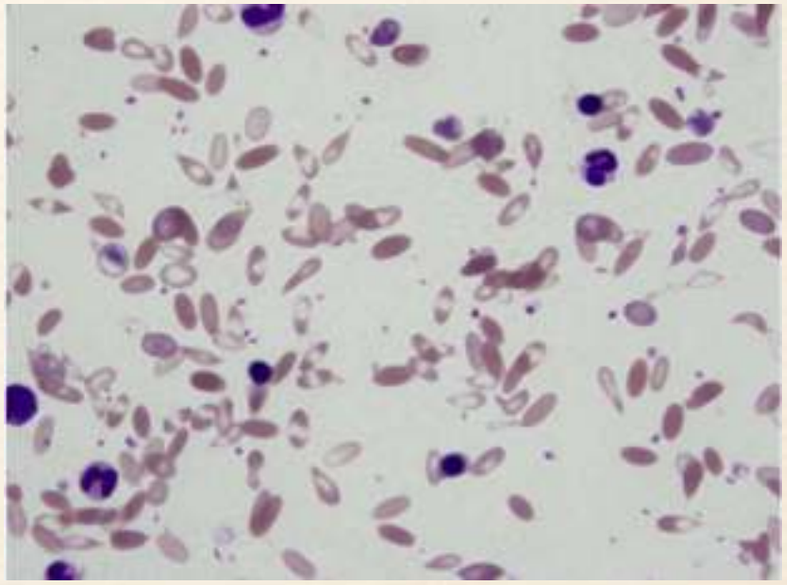
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Hematology

- Alpaca and Llama
  - Morphology
    - Environmental adaptation
    - Hemoglobin crystal
- Anemia
  - Relatively common





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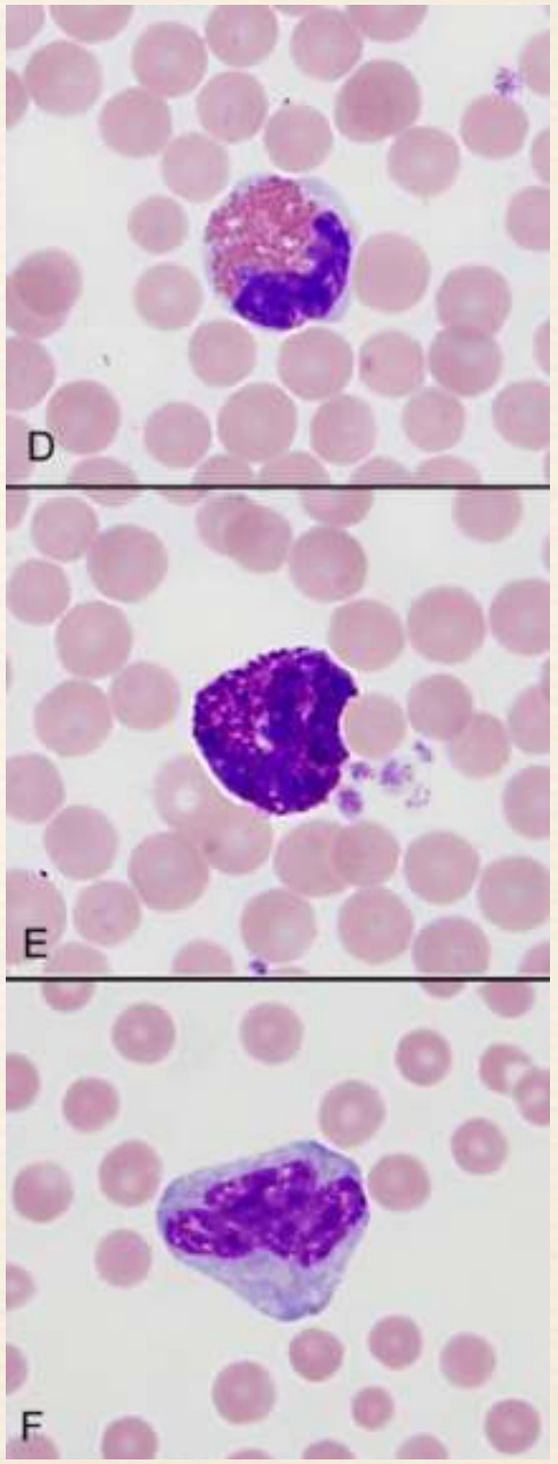
Hematology

- Lymphocyte
  - Outnumber neutrophil when 6-8 month of age
- Lymphocytosis
  - Chronic viral infection
  - Chronic pyogenic disease
  - Autoimmune disease

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Hematology

- Eosinophil
  - Response to parasite
- Basophil
  - Rare in adult
  - Allergic and inflammatory process
    - Histamine
- Monocyte
  - Not a sensitive disease indicator



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Chemistry profile

GENERAL PANEL. The broadly based general chemistry panel should provide a balanced evaluation of the most likely medical problems.

Recommended

Glucose
Blood urea nitrogen (BUN)
Creatinine
Creatine kinase (CK)
Aspartate aminotransferase (AST)
Sorbitol dehydrogenase (SDH)
γ-Glutamyltransferase (GGT)
Alkaline phosphatase (ALP)
Bilirubin (direct, indirect, and total)
Total protein
Albumin
Globulin
Albumin/globulin ratio (A/G ratio)
Bicarbonate or total carbon dioxide (total CO_2)
Sodium
Potassium
Chloride
Calcium
Phosphate

Optional

Venous or arterial blood gases
Ionized calcium

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Chemistry profile

- Glucose
  - Unique metabolism
    - Do not absorb preformed glucose from gut
      - Lower reference range
        - 50-75 mg/dl
        - No postprandial hyperglycemia
- Hyperglycemia
  - Enterotoxemia, Listeriosis
  - Stress
  - DM
    - Rare
- Hypoglycemia
  - Septicemia, malnutrition
  - Pregnancy toxemia

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Chemistry profile

- BUN
  - Same urea cycle as in other mammals
  - BUN is recycled in functional rumen
    - Decrease usefulness in diagnosis in renal disease
- Creatinine
  - Not affected by metabolic process
  - Uroabdomen
    - 1.5 -2 x serum concentration
- BUN & Creatinine
  - 75% function loss
- USG
  - 66% nephron not functional
  - >1.025 pre renal azotemia
  - <1.025 post renal azotemia

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Chemistry profile

- Liver enzyme
  - ALT
    - Low in ruminant
  - AST, LDH, CK
    - Differentiate muscle and liver damage
  - GGT
    - Liver specific
      - Cholestasis, Hepatobiliary disorder
      - Hepatic lipidosis
    - Renal GGT
    - Colostrum GGT
- Vagal indigestion, abomasal volvulus
- Neonate < 1week
  - A battery of tests is indicated

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Chemistry profile

- Protein
  - A/G ratio
    - Easily overlooked
    - Sensitive
    - Chronic inflammatory process
- Hypoglobulinemia
  - Neonate
    - Passive transfer of maternal Ab
    - Rapid consume of transferred Ab

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Lactate

- Lactic acid
  - L- Lactate
    - Produced by mammalian cells
    - Anaerobic metabolism
    - Metabolic acidosis
  - D- Lactate
    - Microbial metabolism
    - Correlate with acidosis in calf diarrhea
- Hyperlactatemia
  - Rumen lactic acidosis (grain overload or toxic indigestion)
  - Hypovolemic shock
  - Septic shock
  - Decreased tissue perfusion

Take home message

- Goat and sheep are different animals from companion mammals
- Respect and understand the similarities and differences will help us make diagnosis
- A GP don't have to remember all ruminant diseases, but need to learn how to " do no harm"